Topic #:-

Basis of vector Space

=> Let $B = \{v_1, v_2, \ldots, v_k\}$ be a set of vectors of a vector space or in a finite-dimensional
=> Then B is called a basis for V if the following two conditions hold:

1. B is linearly independent
2. B spans V.

The vectors $v_1, v_2, \ldots, v_n$ will be called as the basis vectors of V.

How to find basis set:

1) Form the linear combination for the given vector like
$$a_1v_1 + a_2v_2 + \ldots + a_nv_n = 0$$
2) Make augmented matrix from the homogeneous system formed above.
3) Convert augmented matrix to reduced echelon form

The vectors corresponding to the columns containing leading ones form the basis set.

Note:

$e_1 = (1, 0, 0, \ldots, 0)$, $e_2 = (0, 1, 0, \ldots, 0)$, ..., $e_n = (0, 0, \ldots, 0, 1)$ which is called the standard basis for R^n.

In Short:-

=> A linear independent and spanning set of vector space is known as basis of vector space.

=> A vector space can have more than one basis..

Example:

$M_2(F)$ - 2×2 matrices

$e_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$,

$e_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $e_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

Show, $V = \{e_1, e_2, e_3, e_4\}$ is basis of V.

Let consider linear combination

$\alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3 + \alpha_4 e_4 = 0$

$$\Rightarrow \alpha_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \alpha_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \alpha_1 & \alpha_2 \\ \alpha_3 & \alpha_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$$

Let

$$x = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in V, \quad a, b, c, d \in F$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$V = a e_1 + b e_2 + c e_3 + d e_4$

V is a spanning set.

$\therefore$ V is a spanning set

$\therefore$ $x \in V$

$\therefore$ it form basis for vector space for $M_2 F$

Q# Show that (2, 4, -3), (0, 1, 1), (0, 1, -1) from the basis of $\mathbb{R}^3$.

i) Linear independence checking :-

$$k_1V_1 + k_2V_2 + k_3V_3 = 0$$

$$k_1(2, 4, -3) + k_2(0, 1, 1) + k_3(0, 1, -1) = (0, 0$$

$$= \begin{vmatrix} 2 & 0 & 0 \\ 4 & 1 & 1 \\ -3 & 1 & -1 \end{vmatrix}$$

$$= 2\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} - 0\begin{vmatrix} 4 & 1 \\ -3 & -1 \end{vmatrix} + 0\begin{vmatrix} 4 & 1 \\ -3 & 1 \end{vmatrix}$$

$$= 2(-1-1) - 0 + 0$$

$$= 2(-2) = -4 \neq 0$$

So, the given vectors are linearly independent $k_1 = k_2 = k_3 = 0$

ii) Span checking

$$V = aV_1 + bV_2 + cV_3$$

$$(x, y, z) = a(2, 4, -3) + b(0, 1, 1) + c(0, 1, -1)$$

$$\left[\begin{array}{ccc:c} 2 & 0 & 0 & x \\ 4 & 1 & 1 & y \\ -3 & 1 & -1 & z \end{array}\right]$$

$R_3 \to R_3 + R_2$
$$\left[\begin{array}{ccc:c} 2 & 0 & 0 & x \\ 4 & 1 & 1 & y \\ 1 & 2 & 0 & z+y \end{array}\right]$$

$\frac{1}{2} R_1 \to R_1$
$$\left[\begin{array}{ccc:c} 1 & 0 & 0 & x/2 \\ 4 & 1 & 1 & y \\ 1 & 2 & 0 & z+y \end{array}\right]$$

$R_2 \to R_2 - 4R_1$
$$\left[\begin{array}{ccc:c} 1 & 0 & 0 & x/2 \\ 0 & 1 & 1 & y-2x \\ 1 & 2 & 0 & z+y \end{array}\right]$$

$R_3 \to R_3 - R_1$
$$\left[\begin{array}{ccc:c} 1 & 0 & 0 & x/2 \\ 0 & 1 & 1 & y-2x \\ 0 & 2 & 0 & \frac{2z+2y-x}{2} \end{array}\right]$$

$\frac{1}{2} R_3$
$$\left[\begin{array}{ccc:c} 1 & 0 & 0 & x/2 \\ 0 & 1 & 1 & y-2x \\ 0 & 1 & 0 & \frac{2z+2y-x}{4} \end{array}\right]$$

$$\sim R_2 \to R_2 - R_3 \left[\begin{array}{ccc:c} 1 & 0 & 0 & x/2 \\ 0 & 0 & 1 & \frac{2y-7x-2z}{4} \\ 0 & 1 & 0 & \frac{2z+2y-x}{4} \end{array}\right]$$

$a = x/2$

$c = (2y-7x-2z)/4$

$b = (2z+2y-x)/4$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{x}{2}\begin{bmatrix} 2 \\ 4 \\ -3 \end{bmatrix} + \frac{(2y-7x-2z)}{4}\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + \frac{(2z+2y-x)}{4}\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ 2x \\ \frac{-3x}{2} \end{bmatrix} + \begin{bmatrix} 0 \\ (2y-7x-2z)/4 \\ -(2y-7x-2z)/4 \end{bmatrix} + \begin{bmatrix} 0 \\ (2z+2y-x)/4 \\ (2z+2y-x)/4 \end{bmatrix}$$

$$= \begin{bmatrix} x + 0 + 0 \\ 2x + (2y-7x-\cancel{2z}+\cancel{2z}+2y-x)/4 \\ \frac{-3x}{2} + (-2y+7x-2z+2z+2y-x)/4 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

So, the given vectors form the basis of R^3

Q# Show that the vectors $v_1 = (1,2,1)$, $v_2 = (2,9,0)$ and $v_3 = (3,3,4)$ form a basis for R^3

i) for Linear independence

$$v_1k_1 + v_2k_2 + v_3k_3 = 0$$

$$k_1(1,2,1) + k_2(2,9,0) + k_3(3,3,4) = 0$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 9 & 3 \\ 1 & 0 & 4 \end{bmatrix}$$

$$|A| = 1\begin{vmatrix} 9 & 3 \\ 0 & 4 \end{vmatrix} - 2\begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} + 3\begin{vmatrix} 2 & 9 \\ 1 & 0 \end{vmatrix}$$

$$= 1(36-0) - 2(8-3) + 3(0-9)$$

$$= 36 - 10 - 27$$

$$= -1 \neq 0$$

So, $|A| \neq 0$, the system has trivial solution.

$$k_1 = k_2 = k_3 = 0$$

ii) For Span checking:

Let $b = (b_1, b_2, b_3)$

$$k_1v_1 + k_2v_2 + k_3v_3 = b$$

$$(b_1, b_2, b_3) = k_1(1,2,1) + k_2(2,9,0) + k_3(3,3,4)$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 2 & 9 & 3 & b_2 \\ 1 & 0 & 4 & b_3 \end{array}\right]$$

$R_1 \to R_1 - R_3$, $R_2 \to R_2 - 2R_3$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & b_1 - b_3 \\ 0 & 9 & -5 & b_2 - 2b_3 \\ 1 & 0 & 4 & b_3 \end{array}\right]$$

$R_1 \to R_1 - R_2$

$$\left[\begin{array}{ccc|c} 0 & -7 & 4 & b_1 - b_2 + b_3 \\ 0 & 9 & -5 & b_2 - 2b_3 \\ 1 & 0 & 4 & b_3 \end{array}\right]$$

$R_2 \to R_2 + R_1$

$$\left[\begin{array}{ccc|c} 0 & -7 & 4 & b_1 - b_2 + b_3 \\ 0 & 2 & -1 & b_1 - 2b_3/2 \\ 1 & 0 & 4 & b_3 \end{array}\right]$$

$\frac{1}{2}R_2 \to R_2$

$R_1 \to R_1 + 7R_2$

$$\left[\begin{array}{ccc|c} 0 & 0 & 1/2 & 7b_1 - 5b_3 - 2b_2/2 \\ 0 & 1 & -1/2 & (b_1 - 2b_3)/2 \\ 1 & 0 & 4 & b_3 \end{array}\right]$$

(2) $R_1 \to R_1$

$R_2 \to R_2 + R_1$
$$\left[\begin{array}{ccc|c} 0 & 0 & 1/2 & (9b_1 - 5b_3 - 2b_2)/2 \\ 0 & 1 & 0 & \frac{10b_1 - 6b_3 - 2b_2}{2} \\ 1 & 0 & 4 & b_3 \end{array}\right]$$

(2) $R_1 \to R_1$
$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 9b_1 - 5b_3 - 2b_2 \\ 0 & 1 & 0 & (10b_1 - 6b_3 - 2b_2)/2 \\ 1 & 0 & 4 & b_3 \end{array}\right]$$

$R_3 \to R_3 - 4R_1$
$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 9b_1 - 5b_3 - 2b_2 \\ 0 & 1 & 0 & (10b_1 - 6b_3 - 2b_2)/2 \\ 1 & 0 & 0 & 21b_3 - 36b_1 + 8b_2 \end{array}\right]$$

$R_1 \longleftrightarrow R_3$
$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 21b_3 - 36b_1 + 8b_2 \\ 0 & 1 & 0 & (10b_1 - 6b_3 - 2b_2)/2 \\ 0 & 0 & 1 & 9b_1 - 5b_3 - 2b_2 \end{array}\right]$$

$k_1 = 21b_3 - 36b_1 + 8b_2$

$k_2 = (10b_1 - 6b_3 - 2b_2)/2$

$k_3 = 9b_1 - 5b_3 - 2b_2$

Verification.

$(b_1, b_2, b_3) = (21b_3 - 36b_1 + 8b_2)(1,2,1) +$
$(10b_1 - 6b_3 - 2b_2)/2\,(2,9,0) + (9b_1 - 5b_3 - 2b_2)$
$(3,3,4)$

$$(b_1, b_2, b_3) = (21b_3 - 36b_1 + 8b_2)\begin{bmatrix}1\\2\\1\end{bmatrix} +$$

$$(10b_1 - 6b_3 - 2b_2)/2 \begin{bmatrix}2\\9\\0\end{bmatrix} + (9b_1 - 5b_3 - 2b_2)\begin{bmatrix}3\\3\\4\end{bmatrix}$$

$$(b_1, b_2, b_3) = \begin{bmatrix} 27b_1 - 15b_3 - 6b_2 + 10b_1 - 6b_3 - 2b_2 + 21b_3 - 36b_1 + 8b_2 \\ \frac{90b_1 - 54b_3 - 18b_2}{2} + 42b_3 - 72b_1 + 16b_2 + 27b_1 - 15b_3 - 6b_2 \\ 0 + 21b_3 - 36b_1 + 8b_2 + 36b_1 - 20b_3 - 8b_2 \end{bmatrix}$$

$$(b_1, b_2, b_3) = \begin{bmatrix} -\cancel{21b_3} + \cancel{21b_3} + \cancel{8b_2} - \cancel{8b_2} + 37b_1 - 36b_1 \\ 96b_1 - 90b_1 - \cancel{54b_3} + \cancel{54b_3} + 20b_2 - 18b_2/2 \\ -20b_3 + 21b_3 - \cancel{36b_1} + \cancel{36b_1} + \cancel{8b_2} - \cancel{8b_2} \end{bmatrix}$$

$$(b_1, b_2, b_3) = \begin{bmatrix} b_1 \\ 2b_2/2 \\ b_3 \end{bmatrix}$$

$\therefore$ Hence, the given vectors form a basis of R^3

$$(b_1, b_2, b_3) = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$(b_1, b_2, b_3) = (b_1, b_2, b_3)$$

What is finite-dimensional and infinite dimensional?

=> A non-zero vector space V is called finite-dimensional if it contains a finite set of vectors

$$B = \{v_1, v_2, \dots, v_k\}$$

that forms a basis.

=> if no such set exist, V is called infinite-dimensional.

e.g. The zero vector space to be a finite dimensional.

e.g For example, the vector space R^n, P^n and $M_{mn}(F)$ are finite dimensional.

e.g for example, the vector spaces $F(-\infty, \infty)$, $C(-\infty, \infty)$, $C^2(-\infty, \infty)$, $C^n(-\infty, \infty)$ and $C^\infty(-\infty, \infty)$ are infinite dimensional.

Theorem:

⇒ Let V be a finite-dimensional vector space and $\{v_1, v_2, \dots, v_k\}$ be any basis.

- if a set has more than n vectors, then it is linearly dependent.
- if a set has fewer than n vectors, then it does not span V.

Theorem:-

⇒ All basis for a finite-dimensional vector space have the same number of vectors.

Uniqueness of basis representation

⇒ if $S = \{v_1, v_2, \dots, v_n\}$ is a basis for a vector space V, then every vector v in V can be expressed in the form $v = c_1v_1 + c_2v_2 + \dots + c_nv_n$ in exactly one way.

Proof:

Since S spans V, it follows from the definition of spanning set that every vector in V is expressible as a linear combination of the vectors in S.

Suppose that vector v can be written as:

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n \quad \text{—(i)}$$

and

$$v = k_1 v_1 + k_2 v_2 + \cdots k_n v_n \quad \text{—(ii)}$$

Subtract (ii) from (i)

$$0 = (c_1 - k_1) v_1 + (c_2 - k_2) v_2 + \cdots + (c_n - k_n) v_n$$

∴ the right side of this equation is a linear combination of vectors in S, the linear independence of S implies that

$$c_1 - k_1 = 0, \quad c_2 - k_2 = 0, \ldots \quad c_n - k_n = 0$$

$$c_1 = k_1, \quad c_2 = k_2, \ldots, \quad c_n = k_n$$

Order basis:

Definition:
in which the listing order of the basis vectors remains fixed.

Coordinate vector of v relative to S:

⇒ if $S = \{v_1, v_2, \ldots, v_n\}$ is a basis for a vector space V, and

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$$

is the expression for a vector v in terms of the basis S, then the scalars $c_1, c_2, \ldots c_n$ are called the coordinates of v relative to the basis of S.

⇒ The vector $(c_1, c_2, \ldots, c_n)$ in R^n constructed from these coordinates is called the coordinate vector of v relative to S

⇒ it is denoted by $(v)_S = (c_1, c_2, \ldots, c_n)$

Find the coordinate vector for the polynomial

$$P(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$$

relative to the standard basis for the vector space P_n.

$P(x)$ expresses polynomial as a linear combination of the standard basis vectors $S = \{1, x, x^2, \ldots x^n\}$. Thus, the coordinate vector for p relative to S is

$$(P)_S = (c_0, c_1, c_2, \ldots, c_n)$$

Find the coordinate vector of $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ relative to the standard basis for M_{22}.

The representation of a vector as a linear combination of the standard basis vectors is

$$B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} +$$

$$b\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

So, the coordinate vector of B relative to S is

$$(B)_S = (a, b, c, d)$$

Show that the vectors

$V_1 = (1, 2, 1)$

$V_2 = (2, 9, 0)$

$V_3 = (3, 3, 4)$

form the basis of R^3. Find the coordinate vector of $V = (5, -1, 9)$ relative to the basis $S = \{V_1, V_2, V_3\}$

To find $(V)_S$ we should show V as a linear combination of the vectors in S, and find the values of c_1, c_2, c_3

$$V = V_1 c_1 + V_2 c_2 + V_3 c_3$$

, in terms of components

$$(5, -1, 9) = c_1 (1, 2, 1) + c_2 (2, 9, 0) + c_3 (3, 3, 4)$$

Equating corresponding components

$$c_1 + 2c_2 + 3c_3 = 5$$

$$2c_1 + 9c_2 + 3c_3 = -1$$

$$c_1 + 4c_3 = 9$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 2 & 9 & 3 & -1 \\ 1 & 0 & 4 & 9 \end{array}\right]$$

$$R_3 - R_1 \to R_3, \quad R_2 \to R_2 - 2R_1 \quad \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 5 & -3 & -11 \\ 0 & -2 & 1 & 4 \end{array}\right]$$

$$R_1 \to R_1 + R_3 \quad \left[\begin{array}{ccc|c} 1 & 0 & 4 & 9 \\ 0 & 5 & -3 & -11 \\ 0 & -2 & 1 & 4 \end{array}\right]$$

$$R_2 \to R_2 + R_3 \quad \left[\begin{array}{ccc|c} 1 & 0 & 4 & 9 \\ 0 & 3 & -2 & -7 \\ 0 & -2 & 1 & 4 \end{array}\right]$$

$$-\frac{1}{2} R_3 \quad \left[\begin{array}{ccc|c} 1 & 0 & 4 & 9 \\ 0 & 3 & -2 & -7 \\ 0 & 1 & -\frac{1}{2} & -2 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 4 & 9 \\ 0 & 3 & -2 & -7 \\ 0 & 1 & -\frac{1}{2} & -2 \end{array}\right]$$

$$R_2 \to R_2 - 3R_3 \left[\begin{array}{ccc:c} 1 & 0 & 4 & 9 \\ 0 & 0 & -\frac{1}{2} & -1 \\ 0 & 1 & -\frac{1}{2} & -2 \end{array}\right]$$

$$R_3 \to R_3 - R_2 \left[\begin{array}{ccc:c} 1 & 0 & 4 & 9 \\ 0 & 0 & -1/2 & -1 \\ 0 & 1 & 0 & -1 \end{array}\right]$$

$$-(2) R_2 \left[\begin{array}{ccc:c} 1 & 0 & 4 & 9 \\ 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & -1 \end{array}\right]$$

$$R_1 \to R_1 - 4R_2 \left[\begin{array}{ccc:c} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & -1 \end{array}\right]$$

$$R_2 \leftrightarrow R_3 \left[\begin{array}{ccc:c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$c_1 = 1,\ c_2 = -1,\ c_3 = 2$

$V(S) = (1, -1, 2)$

(b) Find the vector v in R^3 whose coordinate vector relative to S is $V(S) = (-1, 3, 2)$

$V = c_1 v_1 + c_2 v_2 + c_3 v_3$

$= (-1)(1,2,1) + 3(2,9,0) + 2(3,3,4)$

$= (11, 31, 7)$